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To

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cc

Jocelyne Modine <jmodine@bio.org>

Subject

Docket ID No. EPA-OAR-2005-0161

Attached are the BIO IES comments for Docket ID No. EPA-OAR-2005-0161

Brent Erickson
Executive Vice President
Biotechnology Industry
Organization (BIO)
1225 Eye St. N.W. S. 400
Washington, D.C. 20005

November 11, 2006-11-07

U.S. Environmental Protection Agency
EPA West (Air Docket)
1200 Pennsylvania Avenue, NW
Room B108 Mail Code 6102 T
Washington, DC 20460
Docket ID No. EPA-OAR-2005-0161

Dear Sir/Madam:

The Biotechnology Industry Organization (BIO) Industrial and Environmental Section (IES) appreciates the opportunity to comment on EPA's Notice of Proposed Rulemaking "Regulation of Fuels and Fuel Additives: Renewable Fuel Standard Program" [Federal Register, Volume 71, Number 184, Friday, September 22, 2006]. BIO is the world's largest biotechnology trade association of companies and laboratories that use biological systems and methods for the production of medical, agricultural and industrial products. BIO represents more than 1,100 members in all 50 states and 31 foreign nations, and has taken an active role in assisting in the development of regulations and policies that affect the biotechnology industry both internationally and in the US. Its membership is global and represents a majority of the US biotechnology industry.

BIO's IES member companies are leaders in the field of industrial biotechnology, ethanol and biodiesel production, novel enzyme biocatalysts, novel processes to make renewable fuels, biobased products and chemicals from renewable feedstocks, and they are revolutionizing the manufacturing of pharmaceuticals, food additives, flavorings, and personal care products.

BIO IES companies are involved in making conventional ethanol as well as ethanol from DDGs and Cellulosic feedstocks. Cellulosic biomass ethanol (CBE) from agricultural residues and dedicated energy crops represents a highly promising option to greatly expand current biofuels production. In fact, a recent report¹ found the land resources in the U.S. are capable of producing a sustainable supply of 1.3 billion tons per year of biomass, and that 1 billion tons of biomass would be sufficient to displace 30 percent or more of the country's present petroleum consumption.

BIO believes biofuels will significantly address U.S. energy problems, and that biotechnology holds the key to driving down the costs of biofuel production throughout the value chain, including the development of new feedstocks, novel enzymes, and fermentation technology. The biotechnology industry supports increased ethanol production from all sources. Biotech can help with ethanol production — from biotech-improved ag feedstocks with improved yield or starch composition, to biotech enzymes that will make current ethanol processes more efficient. Biofuels production ultimately will vary by geography, what a region can produce, and by market opportunities and technology.

¹DOE/USDA Billion ton report

The BIO IES generally supports EPA's proposed regulation to implement the renewable fuels standard (RFS) program authorized in Section 1501 of the Energy Policy Act of 2005, but there are some concerns as outlined in our comments and recommendations below:

The EPACT definition of CBE is as follows:

"The term 'cellulosic biomass ethanol' means ethanol derived from any lignocellulosic or hemicellulosic matter that is available on a renewable or recurring basis, including-

- i. dedicated energy crops and trees;
- ii. wood and wood residues;
- iii. plants;
- iv. grasses;
- v. agricultural residues;
- vi. fibers;
- vii. animal wastes and other waste materials;
- and
- viii. municipal solid waste.

The term also includes any ethanol produced in facilities where animal wastes or other waste materials are digested or otherwise used to displace 90 percent or more of the fossil fuel normally used in the production of ethanol."

In this proposed regulation (71 FR 55568), EPA interprets the statutory language as follows:

"The definition of cellulosic biomass ethanol in the Act also provides that ethanol made at any facility-regardless of whether cellulosic feedstock is used or not-may be defined as cellulosic if at such facility "animal wastes or other waste materials are digested or otherwise used to displace 90 percent or more of the fossil fuel normally used in the production of ethanol." The statutory language suggests that there are two methods through which "animal and other waste materials" may be considered for displacing fossil fuel. The first method is the digestion of animal wastes or other waste materials. EPA proposes to interpret the term "digestion" to mean the conversion of animal or other wastes into methane, which can then be combusted as fuel. We base our interpretation on the practice in industry of using anaerobic digesters to break down waste products such as manure into methane. Anaerobic digestion refers to the breakdown of organic matter by bacteria in the absence of oxygen, and is used to treat waste to produce renewable fuels. We note also that the digestion of animal wastes or other waste materials to produce the fuel used at the ethanol plant does not have to occur at the plant itself. Methane made from animal or other wastes offsite and then purchased and used at the ethanol plant would also qualify. The second method is suggested by the term "otherwise used" which we propose to interpret as meaning (1) the direct combustion of the waste materials as fuel at an ethanol plant, or (2) the use of thermal energy

that itself is a waste product; e.g., waste heat that is obtained from an off-site combustion process such as a neighboring plant that has a furnace or boiler from which the waste heat is captured. With respect to the first meaning, waste materials from tree farms (tops, branches, limbs, etc), or waste materials from saw mills (sawdust, shavings and bark) as well as other vegetative waste materials such as corn stover, or sugar cane bagasse, could be used as fuel for gasifier/boiler units at ethanol plants, since they are waste materials and would not be used as a feedstock to carbohydrate-based ethanol plants. Although such waste materials conceivably could be feedstocks to a cellulosic ethanol plant, its use as a fuel at a carbohydrate based ethanol plant does not subvert the intent of the definition.”

We strongly oppose EPA’s interpretation of the EPACT definition of CBE. This is not the current industry consensus of the currently accepted definition of CBE, which is ethanol produced using lignocellulosic biomass as direct feedstock and enzymatic or acid hydrolysis as a means of producing sugars for fermentation to ethanol. EPA has interpreted the statute overly broadly and there is no legislative history to support such an interpretation.

After a through review of the legislative history, we urge EPA to interpret CBE definition in its strictest possible sense. At a minimum, ethanol produced using waste heat from combustion captured from offsite facilities should be excluded from the definition of CBE. Using waste heat to power starch ethanol production is not technically CBE, since this process does not use lignocellulosic biomass as direct feedstock. We have reviewed the legislative history of the statute and find nothing to support the use of waste heat produced as a result of combustion at offsite facilities. The intent of the statutory language was to facilitate the construction of ethanol plants that can use manure to produce biogas to fuel the boilers at conventional ethanol or CBE plants, or the use of crop residues (such as corn stover) and other cellulosic wastes (such as waste paper) to fire a boiler. A facility that uses ag cellulosic crop residues such as corn stover to fire a boiler and generate energy for existing ethanol units and future CBE production units on the same site is, however, logically included in the statutory definition.

EPA states in the proposed rule, “(2) the use of thermal energy that itself is a waste product; e.g., waste heat that is obtained from an off-site combustion process such as a neighboring plant that has a furnace or boiler from which the waste heat is captured. This interpretation does subvert the intent of the statute. Waste heart from combustion could conceivably come from a coal fired boiler that uses no cellulosic material to generate heat. This interpretation would undermine the intended incentive to foster and promote CBE infrastructure construction and production.

Additionally, EPA mentions in this proposed regulation (71 FR 55565) that prior to 2013, EPACT specifies that ethanol from cellulose will be considered equivalent to 2.5 gallons of renewable fuel when determining compliance with the renewable volume obligation. Beginning in 2013, the 2.5 to 1 ratio will no longer apply for cellulosic biomass ethanol, and in its place, EPACT requires that the volume of required renewable fuel from 2013

and beyond will include a minimum of 250 million gallons that are derived from cellulosic biomass. EPA then concludes that “an obligated party would be subject to two standards in 2013, a non-cellulosic standard and a cellulosic standard”.

According to both this proposed regulation (71 FR 55565) and in EPACT, the cellulosic standard for 2013 and beyond refers to renewable fuels “derived from cellulosic biomass”, not CBE. This standard should therefore apply only to renewable fuels specifically derived from cellulosic biomass feedstocks. Lignocellulosic feedstocks are defined under section 932(a)(2) of EPACT, and are clearly restricted to the commonly accepted definition of “any portion of a plant or coproduct from conversion, including crops, trees, forest residues, and agricultural residues not specifically grown for food, including from barley grain, grapeseed, rice bran, rice hulls, rice straw, soybean matter, and sugarcane bagasse.” In this proposed rulemaking, EPA explicitly did not use the phrase cellulosic biomass ethanol to apply to the cellulosic RFS. The phrase “cellulosic biomass ethanol” is used to establish a 2.5 to 1 equivalence value of renewable fuel when determining the compliance with the non-cellulosic RFS standard prior to 2013.

We believe the legislative intent was that solely ethanol directly derived from cellulosic agricultural or forest feedstocks as defined in EPACT Section 932(a) (2) can qualify towards the cellulosic standard.

- The RFS stipulates that cellulosic biomass ethanol will be considered equivalent to 2.5 gallons of renewable fuel when determining compliance with renewable fuel obligation. This credit will apply until 2013, and in its place, there will be a minimum production requirement of 250 million gallons of renewable fuel derived from cellulosic biomass. The RFS program is centered on credit trading provisions, which will comprise a critical element of compliance. Many obligated parties do not have easy access to renewable fuels, or the ability to blend them, and will rely on the use of credits to comply. Additionally, the refiners, importers, and blenders of gasoline that are liable for meeting this standard are not generally the parties who will be making renewable fuels such as cellulosic biomass ethanol, which creates the need for trading mechanisms that will ensure compliance. Unused credits, as proposed in this regulation, would be valid for 12 months as of the date of generation, and an obligated party can use them to demonstrate compliance, bank them, or transfer them to another party. We believe the unused credits should be valid for 36 months to allow for greater flexibility in the market place.
- As proposed in this regulation, each gallon of cellulosic biomass ethanol would receive 2.5 “credits”, as compared to one gallon of other renewable fuels. In 2013, if the total amount of ethanol produced from cellulosic feedstocks exceeds the 250 million gallon requirement, there is a potential source of additional credits that should be accounted for and utilized in the credit trading program. This would incentivize the production of ethanol derived from cellulosic feedstocks and encourage maximal production under the RFS.

In conclusion, BIO generally supports EPA's efforts in this proposed regulation to implement the Renewable Fuels Standard and to help the United States produce more renewable fuels in the future.

Sincerely,

A handwritten signature in black ink, appearing to read "Brent Er", with a stylized flourish at the end.

Brent Erickson
Vice President, BIO IES